Chapter 7. Matching and Instrumental Variables

Nearest-neighbour matching, Abadie–Imbens inference, IV estimators, compliers and the LATE

Contents

7.1 Matching **3**

7.1.1 Matching as nonparametric imputation . . . 3
7.1.1.1 Setup . . . 3
7.1.1.2 From regression to matching . . . 3
7.1.1.3 The matching idea . . . 3
7.1.1.4 The matching estimator . . . 3
7.1.1.5 Exact matching and coarsened exact matching . . . 4
7.1.2 Nearest-neighbor matching and its theory . . . 4
7.1.2.1 Definition . . . 4
7.1.2.2 Algorithms (matching without replacement) . . . 4
7.1.2.3 The curse of dimensionality . . . 4
7.1.2.4 ATT matching: independence and consistency . . . 4
7.1.2.5 A toy example . . . 5
7.1.2.6 Bias-corrected matching (Abadie and Imbens 2011) . . . 5
7.1.2.7 Inference after matching . . . 5
7.1.2.8 Kernel matching . . . 5
7.1.3 Propensity-score matching . . . 6
7.1.4 Practical matching . . . 6
7.1.5 Model dependence and empirical evidence . . . 7
7.1.6 Appendix: ATE matching: why it is harder . . . 7
7.1.7 Appendix: Why the estimated PS can beat the true PS . . . 7
7.1.8 Appendix: The PS paradox (King and Nielsen 2019) . . . 8
7.1.9 Appendix: PS overfitting and the broader picture . . . 8
7.1.10 Appendix: Beyond binary treatment . . . 8
7.1.11 Appendix: Bias-corrected matching versus post-matching regression . . . 8
7.1.12 Appendix: Stratification . . . 8

7.2 Instrumental variables **9**

7.2.1 Motivation . . . 9
7.2.1.1 Instrumental variables in action . . . 9
7.2.1.2 Why instruments? . . . 9
7.2.1.3 Conditions for a good instrument . . . 9
7.2.2 IV under linearity . . . 9
7.2.2.1 The econometric setup . . . 9
7.2.2.2 Just-identified, no covariates . . . 10
7.2.2.3 Just-identified, with covariates . . . 10
7.2.2.4 From linear to nonparametric IV . . . 10
7.2.3 Randomized experiments with non-compliance . . . 11
7.2.3.1 Setting . . . 11
7.2.3.2 Intent-to-treat effects . . . 11
7.2.3.3 Identification assumptions . . . 11
7.2.3.4 Identification . . . 11
7.2.3.5 A worked example . . . 12
7.2.3.6 CACE estimator . . . 12
7.2.3.7 The “fourth condition”: homogeneity . . . 13
7.2.3.8 Summary . . . 13
7.2.4 Appendix: Candidate instruments in observational studies . . . 13

7.2.5 Appendix: IV versus other causal methods 13
7.2.6 Appendix: Practical issues with classic IV studies 13
7.2.7 Appendix: Weak instruments 14
7.2.8 Appendix: Many instruments and robust IV 14

7.1 Matching

Motivation. How to estimate the ATT by imputing each treated unit's missing control outcome from similar controls, with little dependence on modeling choices.

7.1.1 Matching as nonparametric imputation

Motivation. Matching imputes counterfactual control outcomes without a model.

7.1.1.1 Setup

N units, treatment $X_i \in \{0,1\}$, covariates Z_i, potential outcomes $Y_i(0), Y_i(1)$; $N_1 = |\{i : X_i = 1\}|$, $N_0 = N - N_1$. Assume ignorability $(Y(0), Y(1)) \perp\!\!\!\perp X \mid Z$ and positivity $0 < P(X = 1 \mid Z) < 1$. Target: the ATT.

7.1.1.2 From regression to matching

With $\mu_t(Z) = \mathbb{E}[Y \mid X = t, Z]$,

$$\text{ATT} = \mathbb{E}_{Z|X=1}\big[\underbrace{\mu_1(Z)}_{\text{observed}} - \underbrace{\mu_0(Z)}_{\text{impute}}\big].$$

Regression imputes $\mu_0(Z_i)$ by a model:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{\mu}_0(Z_i)).$$

If the groups differ in Z, the result depends on the form of $\hat{\mu}_0$ (Section 7.1.5); matching imputes *nonparametrically*.

7.1.1.3 The matching idea

Given a large reservoir of controls, match each treated unit to one or more controls with similar covariates.

Matching as study design. Matching can be done before outcomes are measured: select treated units and controls, then measure outcomes. Design (whom to compare) is thus separated from analysis and made blind to outcomes (Stuart 2010, §1).

Bias–variance trade-off in M. With M controls per treated unit: $M = 1$ gives lower bias, higher variance; $M = 3$ uses farther matches (more bias) but averages three outcomes (less variance) (Abadie and Imbens 2006).

7.1.1.4 The matching estimator

With matched set $\mathcal{M}_i$ of treated unit i, under ignorability

$$\hat{\tau}_{\text{match}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i\Big(Y_i - \underbrace{\frac{1}{|\mathcal{M}_i|}\sum_{j\in\mathcal{M}_i} Y_j}_{\approx Y_i(0)}\Big).$$

Matching may be one-to-one or one-to-many, with or without replacement.

7.1.1.5 Exact matching and coarsened exact matching

Exact matching: match i to a control with $Z_j = Z_i$; perfect balance, no model dependence, but infeasible with many or continuous covariates.
Coarsened exact matching (CEM; Iacus et al. 2012): discretize covariates, then match exactly on the coarsened values; some treated units may remain unmatched (no overlap).

7.1.2 Nearest-neighbor matching and its theory

Motivation. NN matching: definition, algorithms and large-sample behavior.

7.1.2.1 Definition

Let $\mathcal{M}_i$ index the M closest units of the opposite group in a distance $\|\cdot\|$, and

$$\hat{Y}_i(t) = \begin{cases} Y_i, & X_i = t, \\ \dfrac{1}{M}\sum_{j\in\mathcal{M}_i} Y_j, & X_i = 1-t, \end{cases} \qquad t \in \{0,1\},$$

with $\hat{\tau}_{\text{ATT}} = \frac{1}{N_1}\sum_{i=1}^{N} X_i(Y_i - \hat{Y}_i(0))$.

7.1.2.2 Algorithms (matching without replacement)

- **Greedy:** order the treated units (e.g. hardest first); give each the M closest remaining controls, remove them, repeat.
- **Optimal:** in the $N_1 \times N_0$ distance matrix D choose exactly M entries per row and at most one per column, minimizing total distance (Hungarian algorithm, polynomial time).

Why order matters. With $M = 1$, an arbitrary order lets i_2 take the only control near the isolated i_3 (distance 0.9), forcing i_3 to a control 2.9 away. Hardest-first: i_3 gets its control (1.2), i_2 another (1.1); the worst distance drops from 2.9 to 1.2.

7.1.2.3 The curse of dimensionality

N uniform points in p dimensions have spacing about $N^{-1/p}$ per axis, so the NN distance is $O(N^{-1/p})$. If μ_0 is L-Lipschitz, the mismatch bias satisfies

$$|\mu_0(Z_i) - \mu_0(Z_j)| \le L\,\|Z_i - Z_j\| = O(N^{-1/p}).$$

p	1	5	10	20
$N^{-1/p}$ for $N = 10{,}000$	0.0001	0.16	0.40	0.63

Hence dimension reduction, e.g. propensity-score matching (Abadie and Imbens 2006, Thm. 1).

7.1.2.4 ATT matching: independence and consistency

With fixed M and replacement, each treated unit appears once as a subject, so the imputed $\hat{Y}_i(0)$ are independent across treated units. With the LLN and negligible bias (large control reservoir), the ATT estimator is $\sqrt{N}$-consistent; matching is most natural for the ATT (Abadie and Imbens 2006, Thms. 1–4).

7.1.2.5 A toy example

$N_1 = 3$, $N_0 = 6$, $p = 2$, $M = 1$, distance $\|\cdot\|_1$:

	Treated				Control		
Unit	Z_1	Z_2	Y	Unit	Z_1	Z_2	Y
$i=1$	3	5	8	$j=1$	1	2	2
$i=2$	6	2	12	$j=2$	2	6	4
$i=3$	7	7	10	$j=3$	4	1	6
				$j=4$	3	4	5
				$j=5$	5	3	7
				$j=6$	8	6	9

Matches: $i=1 \to j=4$ (distance 1), $i=2 \to j=5$ (2), $i=3 \to j=6$ (2), so

$$\hat{\tau}_{\text{ATT}} = \tfrac{1}{3}\big[(8-5) + (12-7) + (10-9)\big] = 3.0.$$

The matches are imperfect ($Z_2 - Z_5 \neq 0$); the resulting bias is corrected next.

7.1.2.6 Bias-corrected matching (Abadie and Imbens 2011)

Y_j estimates $\mu_0(Z_j)$, not $\mu_0(Z_i)$; shift it by the predicted mismatch:

$$\hat{\tau}_{\text{bc}} = \frac{1}{N_1} \sum_{i:X_i=1} \Big(Y_i - \frac{1}{M} \sum_{j \in \mathcal{M}_i} \big[Y_j + \hat{\mu}_0(Z_i) - \hat{\mu}_0(Z_j) \big] \Big),$$

with $\hat{\mu}_0$ a regression fit of $\mathbb{E}[Y \mid Z, X=0]$. Default in the R packages `MatchIt` and `Matching`.

7.1.2.7 Inference after matching

ATT variance estimator (Abadie and Imbens 2006):

$$\widehat{\text{Var}}(\hat{\tau}_{\text{ATT}}) = \frac{1}{N_1^2} \sum_{i=1}^{N} \big(1 + K_M(i)\big)^2 \hat{\sigma}_i^2,$$

$K_M(i)$ the number of times i is used as a match, $\hat{\sigma}_i^2$ an estimate of $\text{Var}(Y_i \mid Z_i, X_i)$. Frequently used units inflate the variance, like extreme IPW weights. **The bootstrap is invalid** for NN matching: the matched unit's identity is a discontinuous function of the data (Abadie and Imbens 2008).

7.1.2.8 Kernel matching

All controls within bandwidth h get weight, closer ones more; with kernel K (e.g. $K(u) = e^{-u^2/2}$),

$$w_i(j) = \frac{K\Big(\frac{Z_j - Z_i}{h}\Big)}{\sum_{k:X_k=0} K\Big(\frac{Z_k - Z_i}{h}\Big)}, \qquad \hat{\tau}_{\text{kernel}} = \frac{1}{N_1} \sum_{i=1}^{N} X_i \Big(Y_i - \sum_{j:X_j=0} w_i(j) Y_j \Big).$$

It suffers the same curse of dimensionality.

7.1.3 Propensity-score matching

Motivation. Reduce matching to one dimension, with caveats.

Dimension reduction. With 20 binary covariates there are $2^{20} \approx 10^6$ patterns. The propensity score $e(Z) = P(X = 1 \mid Z)$ is one-dimensional, and $Y(0), Y(1) \perp\!\!\!\perp X \mid Z$ implies $Y(0), Y(1) \perp\!\!\!\perp X \mid e(Z)$.

Distance. $d(Z_i, Z_j) = |\hat{e}(Z_i) - \hat{e}(Z_j)|$, or (preferred, as it stretches differences near 0 and 1)

$$d(Z_i, Z_j) = |\operatorname{logit} \hat{e}(Z_i) - \operatorname{logit} \hat{e}(Z_j)|, \qquad \operatorname{logit}(p) = \ln \frac{p}{1-p}.$$

Specification: balance, not fit. Estimate the PS by logistic regression, GBM, random forest, etc.; the goal is covariate balance, not prediction. Collinearity is irrelevant and overfitting can help (Appendix). Iterate: check balance on covariates, squares and interactions, add terms if imbalanced. Impute missing covariates and add missingness indicators (Stuart 2010; Rosenbaum and Rubin 1984).

Common support. Trim units outside the overlap of the two PS distributions.

Caveats. Matching on the *estimated* PS can be more efficient than on the true PS under correct specification (Abadie and Imbens 2016). The "PS paradox" (King and Nielsen 2019): a narrower caliper improves PS balance but can worsen covariate balance. Always check multivariate covariate balance.

7.1.4 Practical matching

Motivation. The iterative matching workflow and its design choices.

Workflow. Distance → match → check balance → if imbalanced, change metric, M or caliper and repeat; else estimate and infer. Regression or weighting on the matched sample corrects residual imbalance.

Design choices.

- Covariates (adjustment criterion; never post-treatment variables).
- Distance: Mahalanobis, propensity score, or hybrid.
- M fixed or caliper-based; smaller M: less bias, more variance. A large control group suits 1-to-many matching.
- With or without replacement.

Mahalanobis distance $d(x, y) = \sqrt{(x-y)^\top S^{-1}(x-y)}$, S the sample covariance (treated or pooled), works for few continuous covariates ($p \lesssim 5$–8); otherwise the PS reduces dimension. Common practice: PS caliper, then Mahalanobis within it.

Caliper matching. All controls within distance $< c$ are matches, so M varies by unit (e.g. $M_1 = 2$, $M_2 = 3$, $M_3 = 1$). Common choice $c = 0.2 \times \mathrm{SD}(\operatorname{logit} \hat{e}(Z))$ (Austin 2011).

Full matching (Rosenbaum 1991; Hansen 2004). Every unit is placed in a matched set with variable ratio (1:k or k:1), minimizing total within-set distance. It discards no controls in the overlap and yields a weighted sample, between matching and IPW; often with a caliper. R: `MatchIt`, `method = "full"`.

With versus without replacement. With: easier, order-free, but units with extreme PS may be reused and dominate. Without: no reuse, but order-dependent (greedy vs. optimal). Software: `MatchIt` (Ho et al.), `Matching` (Sekhon).

Checking balance. Standardized mean difference of covariate k:

$$\mathrm{SMD}_k = \frac{1}{s_t} \cdot \frac{1}{N_1} \sum_{i=1}^{N} X_i \Big(Z_{ik} - \frac{1}{|\mathcal{M}_i|} \sum_{j \in \mathcal{M}_i} Z_{jk} \Big),$$

s_t the pre-matching treated-group SD. Compare distributions by empirical CDFs and Kolmogorov–Smirnov statistics. **Do not use p-values**: they mix balance with sample size, and balance is an in-sample property (Ho et al. 2007).

Love plot. Dot plot of SMDs per covariate before and after adjustment, lines at ± 0.1. In the LaLonde (1986) data (PS, age, education, married, no degree, race, 1974/1975 earnings), unadjusted |SMD| reach 1.5–2; after matching all lie within ± 0.1.

7.1.5 Model dependence and empirical evidence

Motivation. Matching reduces sensitivity to modeling assumptions; the LaLonde benchmark tests it.

The extrapolation problem. When groups differ on covariates, regression extrapolates into sparse regions: linear and quadratic fits agreeing on the controls can differ widely on the treated (Ho, Imai, King and Stuart 2007).

Matching as preprocessing. After matching, regression interpolates, and specifications agree:

(1) match for balance, (2) regress/weight on the matched sample, (3) obtain estimates robust to model choice.

The LaLonde benchmark. The NSW job-training experiment (LaLonde 1986) has experimental ATT $\approx$ \$1,794. With observational controls (CPS/PSID), which differ sharply in demographics and prior earnings, unadjusted estimates range from $-$\$15,000 to $+$\$2,000.

Lessons. Dehejia and Wahba (1999): PS matching on a subsample with pre-treatment earnings recovers the benchmark. Smith and Todd (2005): fragile across subsamples, methods and metrics. Without pre-treatment earnings, the key confounder, ignorability itself is doubtful. Covariate sufficiency comes before method choice.

Practical takeaways.

- Matching is nonparametric preprocessing; follow it by regression or weighting.
- Check multivariate covariate balance, not just PS balance.
- In high dimensions combine matching with regression.
- Complement with sensitivity analysis for unmeasured confounding.

7.1.6 Appendix: ATE matching: why it is harder

The ATE needs $\hat{Y}_i(0)$ for treated and $\hat{Y}_i(1)$ for controls, so each unit is both subject and match, creating cross-group dependence. For $p > 2$, $\mathrm{Bias}(\hat{\tau}_{\mathrm{ATE}}) = O(N^{-1/p}) \gg O(N^{-1/2})$, so simple matching is not $\sqrt{N}$-consistent; use bias correction (Abadie and Imbens 2011) or IPW/DR (Abadie and Imbens 2006).

7.1.7 Appendix: Why the estimated PS can beat the true PS

Units with $e(Z_i) = e(Z_j)$ can have very different covariates. The fitted $\hat{e}(Z) = g(Z^\top \hat{\beta})$ adapts to the sample, so $\hat{e}(Z_i) \approx \hat{e}(Z_j)$ means closeness in the directions separating treated from control here: implicit covariate refinement. Under correct specification $\mathrm{Var}(\hat{\tau}_{\mathrm{ATT}} \mid \hat{e}) \leq \mathrm{Var}(\hat{\tau}_{\mathrm{ATT}} \mid e)$ (Abadie

and Imbens 2016); similarly IPW with $\hat{e}$ attains the efficiency bound and with e does not (Hirano, Imbens and Ridder 2003). The advantage disappears under misspecification.

7.1.8 Appendix: The PS paradox (King and Nielsen 2019)

Treated unit A (high income, low education) and control B (low income, high education) may both have $\hat{e} = 0.50$: perfect PS balance, opposite covariates. A tighter caliper yields more such pairs, so covariate balance can worsen, requiring more extrapolation than the raw data.

7.1.9 Appendix: PS overfitting and the broader picture

Overfitting can help, since the goal is in-sample balance; the limit is overlap ($\hat{e} \to 0$ or 1). The estimated PS beats the true PS and overfitting helps by the same sample adaptation, yet the paradox shows 1D compression loses information. Use the PS for dimension reduction, always verify multivariate balance, and consider Mahalanobis matching within a PS caliper.

7.1.10 Appendix: Beyond binary treatment

For multiple levels, compare pairs by binary matching or use the generalized propensity score (GPS; Imai and van Dyk 2004). For continuous treatment the GPS is the conditional density of treatment given covariates (Hirano and Imbens 2004); subclassification or weighting on it is more common than matching.

7.1.11 Appendix: Bias-corrected matching versus post-matching regression

Bias correction: matching is primary, regression corrects the $Z_i \neq Z_j$ gap within pairs. Post-matching regression: matching is preprocessing and $Y \sim X + Z$ on the matched data estimates. Full-sample regression uses all controls and is fine if the model is correct, but extrapolates if it is wrong.

	Model correct	Model wrong
Full-sample regression	optimal (more data)	risky (extrapolation)
Post-matching regression	less efficient (data discarded)	safe (interpolation)

Example: controls in $Z \in [0, 3]$, treated in $[5, 8]$; linear and quadratic fits give different $\hat{\mu}_0(6)$, but after matching both agree (Ho et al. 2007).

7.1.12 Appendix: Stratification

For a discrete covariate Z (levels k) making assignment ignorable, $\mathbb{E}[Y(1)] = \sum_k \mathbb{E}(Y \mid Z = k, X = 1)P(Z = k)$, estimated by $\sum_k \bar{Y}_{k,1} n_k/n$ (n_k units, cell means $\bar{Y}_{k,x}$); hence

$$\hat{\tau} = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) \frac{n_k}{n}.$$

For continuous Z split into k classes: $\hat{\tau}_k = \sum_k (\bar{Y}_{k,1} - \bar{Y}_{k,0}) n_k/n$, generally biased. With $R_k = 1 - \frac{\mathbb{E}(\hat{\tau}_k) - \tau}{\mathbb{E}(\hat{\tau}_1) - \tau}$ the fraction of bias removed, Cochran (1968): $R_k \geq 90\%$ for $k \geq 5$ over a broad class of models.

7.2 Instrumental variables

Motivation. How to learn a causal effect under unmeasured confounding from a variable that moves the treatment but affects the outcome only through it.

7.2.1 Motivation

Motivation. Instruments in practice, and what makes one good.

7.2.1.1 Instrumental variables in action

Study	Instrument Z	Treatment X	Outcome Y
Angrist and Krueger (1991)	quarter of birth	years of schooling	earnings
Card (1995)	proximity to college	years of schooling	earnings
Acemoglu et al. (2001)	settler mortality	institutional quality	GDP per capita
Miguel et al. (2004)	rainfall shocks	economic growth	civil conflict
Levitt (1996)	overcrowding litigation	prison population	crime rate

Returns to education. Ability U confounds schooling X and earnings Y. Angrist–Krueger: under compulsory-schooling laws, those born early in the year can drop out sooner. Card: living near a four-year college lowers the cost of attending.
Institutions and development. Geography and culture U confound institutions X and GDP Y. Acemoglu–Johnson–Robinson: high settler mortality led to extractive, persistent institutions. Miguel et al.: rainfall instruments growth in studying African civil conflict.

7.2.1.2 Why instruments?

Back-door adjustment needs all confounders measured. Otherwise, use Z affecting Y only through X ($Z \to X \to Y$, $X \leftarrow U \to Y$, $Z \perp\!\!\!\perp U$): an *instrumental variable*.

7.2.1.3 Conditions for a good instrument

1. **Relevance:** large $\mathrm{Cov}(X, Z)$, else the IV estimate has large variance;
2. **Exclusion restriction:** no direct effect on the outcome;

and (implicitly) no confounding between Z and Y.

7.2.2 IV under linearity

Motivation. Classical IV estimators in the linear constant-effect model.

7.2.2.1 The econometric setup

$Y = \beta_0 + \beta_1 X + \beta_2 W + \varepsilon$, scalar treatment X, covariates W, estimand β_1. X is *endogenous* (correlated with ε), W *exogenous*, so OLS of β_1 is biased. An instrument is a K-vector Z with $\mathrm{Cov}[\varepsilon, Z] = \mathrm{Cov}[\varepsilon, W] = 0$; $K > 1$: over-identified, $K = 1$: just-identified.

7.2.2.2 Just-identified, no covariates

IV estimator.

$$\hat{\beta}_{1,\text{iv}} = \frac{\widehat{\text{Cov}}(Y_i, Z_i)}{\widehat{\text{Cov}}(X_i, Z_i)} = \frac{\frac{1}{N}\sum_{i=1}^{N}(Y_i - \bar{Y})(Z_i - \bar{Z})}{\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})(Z_i - \bar{Z})}.$$

For binary Z, since $\widehat{\text{Cov}}(Y, Z) = \hat{p}(1-\hat{p})(\bar{Y}_1 - \bar{Y}_0)$ ($\hat{p}$ the fraction with $Z = 1$) and likewise for X, this is the *Wald estimator*

$$\hat{\beta}_{1,\text{iv}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0},$$

$\bar{Y}_z, \bar{X}_z$ the means in group $Z = z$.

Indirect least squares (ILS). Fit the reduced forms

$$Y = \pi_{10} + \pi_{11} Z + \varepsilon_1, \qquad X = \pi_{20} + \pi_{21} Z + \varepsilon_2;$$

each slope is $\widehat{\text{Cov}}(\cdot, Z)/\widehat{\text{Var}}(Z)$, so $\hat{\beta}_{1,\text{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21} = \hat{\beta}_{1,\text{iv}}$.

Two-stage least squares (2SLS). Stage 1: $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z$; stage 2: OLS of $Y = \beta_0 + \beta_1 \hat{X} + \varepsilon$. ($\hat{X}$ is the "unconfounded portion" of X.) Since $\widehat{\text{Cov}}(Y, \hat{X}) = \hat{\pi}_{21}\widehat{\text{Cov}}(Y, Z)$ and $\widehat{\text{Var}}(\hat{X}) = \hat{\pi}_{21}^2 \widehat{\text{Var}}(Z)$, $\hat{\beta}_{1,\text{2sls}} = \widehat{\text{Cov}}(Y, Z)/(\hat{\pi}_{21}\widehat{\text{Var}}(Z)) = \hat{\beta}_{1,\text{iv}}$. Hence $\hat{\beta}_{1,\text{iv}} = \hat{\beta}_{1,\text{ils}} = \hat{\beta}_{1,\text{2sls}}$.

7.2.2.3 Just-identified, with covariates

ILS: $Y = \pi_{10} + \pi_{11} Z + \pi_{12} W + \varepsilon_1$, $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$, $\hat{\beta}_{1,\text{ils}} = \hat{\pi}_{11}/\hat{\pi}_{21}$. 2SLS: $\hat{X} = \hat{\pi}_{20} + \hat{\pi}_{21} Z + \hat{\pi}_{22} W$, then $Y = \beta_0 + \beta_1 \hat{X} + \beta_2 W + \varepsilon$; $\hat{\beta}_{1,\text{2sls}}$ is the OLS coefficient of $\hat{X}$.

IV derivation. Let $X = \pi_{20} + \pi_{21} Z + \pi_{22} W + \varepsilon_2$, $Y = \beta_0 + \beta_1 X + \underbrace{\beta_2 W + \varepsilon}_{\eta}$, and assume $\text{Cov}[W, Z] = 0$. Then $\text{Cov}[X, Z] \neq 0$, $\text{Cov}[\eta, Z] = 0$, so

$$\text{Cov}[Y, Z] = \text{Cov}[\beta_0 + \beta_1 X + \eta, Z] = \beta_1 \,\text{Cov}[X, Z],$$

$$\beta_1 = \frac{\text{Cov}[Y, Z]}{\text{Cov}[X, Z]} = \frac{\text{Cov}[Y, Z]/\,\text{Var}[Z]}{\text{Cov}[X, Z]/\,\text{Var}[Z]} = \frac{\text{reduced form}}{\text{first stage}}.$$

2SLS derivation. Substituting X,

$$\begin{aligned} Y &= \beta_1(\pi_{22} W + \pi_{21} Z + \pi_{20} + \varepsilon_2) + \beta_2 W + \beta_0 + \varepsilon \\ &= \beta_1 \pi_{21} Z + (\beta_2 + \beta_1 \pi_{22}) W + \underbrace{(\beta_1(\pi_{20} + \varepsilon_2) + \beta_0 + \varepsilon)}_{\xi}, \end{aligned}$$

hence

$$Y = \beta_1(\pi_{22} W + \pi_{21} Z) + \beta_2 W + \xi = \beta_1 \underbrace{\mathbb{E}[X \mid W, Z]}_{\text{population analogue of } \hat{X}} + \beta_2 W + \xi,$$

with π_{20} absorbed into ξ ($\mathbb{E}[X \mid W, Z] - \pi_{20} = \pi_{21} Z + \pi_{22} W$). Regressing Y on $\mathbb{E}[X \mid W, Z]$ and W recovers β_1; the sample version uses $\hat{X}$.

7.2.2.4 From linear to nonparametric IV

With a constant effect β_1, IV = ILS = 2SLS. What does IV identify when $Y_i(1) - Y_i(0)$ varies? Key case: a randomized experiment with non-compliance, assignment Z instrumenting treatment received X (Angrist, Imbens and Rubin 1996).

7.2.3 Randomized experiments with non-compliance

Motivation. Identify the complier effect under random assignment, monotonicity and exclusion.

7.2.3.1 Setting

RCT of a drug: Z = assignment, X = treatment received, Y = outcome. Some assigned patients refuse; some controls obtain the drug. Comparing $X = 1$ with $X = 0$ is confounded by self-selected compliance; Z is randomized and acts only through X, so it is an instrument.

Potential outcomes (Angrist, Imbens and Rubin 1996). Outcomes $Y(z)$ and treatment received $X(z)$ are indexed by assignment $z = 0, 1$; observed Z, $X = X(Z)$, $Y = Y(Z)$. Latent *compliance types* $S = (X(0), X(1))$:

	$X(0) = 0$	$X(0) = 1$
$X(1) = 0$	never-taker ($X = 0$ always)	defier (X opposes Z)
$X(1) = 1$	complier (X follows Z)	always-taker ($X = 1$ always)

Each observed (Z, X) cell is a mixture: $(0,0)$: {C, NT}; $(0,1)$: {AT, D}; $(1,0)$: {NT, D}; $(1,1)$: {C, AT}; extra assumptions are needed. Goal: $\text{CACE} = \mathbb{E}[Y(1) - Y(0) \mid \text{complier}] = \text{ITT}_y/\text{ITT}_x$.

7.2.3.2 Intent-to-treat effects

S is a baseline characteristic. With $\text{ITT}_s = \mathbb{E}[Y(1) - Y(0) \mid S = s]$, $s \in \{c, n, a, d\}$, and proportions π_s, the global ITT $\text{ITT}_y := \mathbb{E}[Y(1) - Y(0)]$ is

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \text{ITT}_n + \pi_a \text{ITT}_a + \pi_d \text{ITT}_d.$$

7.2.3.3 Identification assumptions

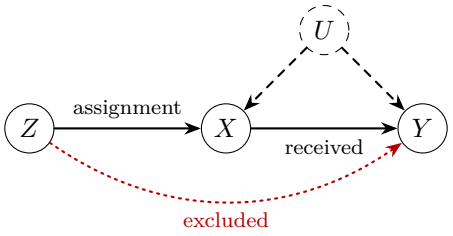


Figure. *Randomized encouragement with non-compliance: Z randomized (A1), no defiers (A2), no direct $Z \to Y$ path (exclusion, A3: dotted arrow absent); U confounds X and Y.*

A1 (random assignment) $\{Y(0), Y(1), X(0), X(1)\} \perp\!\!\!\perp Z$.

A2 (strong monotonicity) (1) no defiers: $X(1) \geq X(0)$ for all units;

(2) compliers exist: $\mathbb{E}[X \mid Z = 1] \neq \mathbb{E}[X \mid Z = 0]$. Often true by design; one-sided compliance means $X(0) = 0$.

A3 (exclusion restriction for non-compliers) $Y(0) = Y(1)$ for all units with $S \in \{n, a\}$: Z has no direct effect (e.g. placebo) on Y, as in linear IV.

7.2.3.4 Identification

A2 gives $\pi_d = 0$; A3 gives $\text{ITT}_n = \text{ITT}_a = 0$. Hence

$$\text{ITT}_y = \pi_c \text{ITT}_c + \pi_n \cdot 0 + \pi_a \cdot 0 + 0 \cdot \text{ITT}_d = \pi_c \text{ITT}_c, \qquad \text{ITT}_c = \text{ITT}_y/\pi_c.$$

With $0 < \pi_c < 1$, $|\text{ITT}_y| < |\text{ITT}_c|$: the global ITT is attenuated.

Compliance-type proportions. Under A1–A2, $X = 1$ at $Z = 0$ only for always-takers and $X = 0$ at $Z = 1$ only for never-takers:

$$\pi_a = P(X = 1 \mid Z = 0) = \mathbb{E}[X \mid Z = 0], \qquad \pi_n = P(X = 0 \mid Z = 1) = 1 - \mathbb{E}[X \mid Z = 1],$$
$$\pi_c = 1 - \pi_a - \pi_n = \mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0].$$

Observed cells under monotonicity. $(0,0)$: compliers and never-takers; $(0,1)$: always-takers; $(1,0)$: never-takers; $(1,1)$: compliers and always-takers. The pure cells identify non-complier means; with the known proportions the mixed cells give the complier means:

$$\mathbb{E}[Y \mid X = 0, Z = 0] = \tfrac{\pi_c}{\pi_c+\pi_n}\mathbb{E}[Y(0) \mid \text{complier}] + \tfrac{\pi_n}{\pi_c+\pi_n}\mathbb{E}[Y(0) \mid \text{never-taker}],$$
$$\mathbb{E}[Y \mid X = 0, Z = 1] = \mathbb{E}[Y(0) \mid \text{never-taker}], \qquad \mathbb{E}[Y \mid X = 1, Z = 0] = \mathbb{E}[Y(1) \mid \text{always-taker}],$$
$$\mathbb{E}[Y \mid X = 1, Z = 1] = \tfrac{\pi_c}{\pi_c+\pi_a}\mathbb{E}[Y(1) \mid \text{complier}] + \tfrac{\pi_a}{\pi_c+\pi_a}\mathbb{E}[Y(1) \mid \text{always-taker}].$$

(Here $Y(0)$, $Y(1)$ are outcomes under treatment received; for compliers they equal the assignment-indexed ones.)

ITT on treatment. $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = \pi_c$: under monotonicity only compliers switch.

Theorem 7.1 (CACE / LATE; Imbens and Angrist 1994, Angrist et al. 1996)

Under A1–A3,

$$\text{CACE} \equiv \mathbb{E}[Y(1) - Y(0) \mid S = c] = \frac{\text{ITT}_y}{\text{ITT}_x} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}.$$

Proof [Supplement]. By A1, $\text{ITT}_y = \mathbb{E}[Y(1, X(1)) - Y(0, X(0))]$; split by $S \in \{c, a, n, d\}$. Always- and never-takers have $X(1) = X(0)$, so by exclusion (A3) they contribute zero; by monotonicity (A2) there are no defiers. Hence $\text{ITT}_y = P(S = c)\,\mathbb{E}[Y(1) - Y(0) \mid S = c]$. Also $\text{ITT}_x = \mathbb{E}[X(1) - X(0)] = P(S = c)$, as $X(1) - X(0)$ is 1 for compliers and 0 otherwise. Dividing $(P(S = c) > 0)$ gives $\text{CACE} = \text{ITT}_y/\text{ITT}_x$, and by A1 each ITT is a difference of observed means. ■

The CACE (LATE) equals the Wald estimand with assignment as instrument.

7.2.3.5 A worked example

$N = 1{,}000$, 500 with $Z = 1$ and 500 with $Z = 0$.

1. **Proportions.** 100 of the $Z = 0$ group take the drug: $\hat{\pi}_a = 0.20$; 150 of the $Z = 1$ group refuse: $\hat{\pi}_n = 0.30$; $\hat{\pi}_c = 0.50$.
2. **ITT effects.** $\bar{Y}_{Z=1} = 7.0$, $\bar{Y}_{Z=0} = 6.0$: $\text{ITT}_y = 1.0$; $\bar{X}_{Z=1} = 0.70$, $\bar{X}_{Z=0} = 0.20$: $\text{ITT}_x = 0.50 = \hat{\pi}_c$.
3. **CACE** $= 1.0/0.50 = 2.0$.

The ITT (1.0) is attenuated; the naive $X = 1$ vs. $X = 0$ comparison is confounded.

7.2.3.6 CACE estimator

For two-sided non-compliance,

$$\widehat{\text{CACE}} = \frac{\widehat{\text{ITT}}_y}{\hat{\pi}_c} = \frac{\sum_i Y_i Z_i / \sum_i Z_i - \sum_i Y_i(1 - Z_i) / \sum_i (1 - Z_i)}{1 - \sum_i X_i(1 - Z_i) / \sum_i (1 - Z_i) - \sum_i (1 - X_i) Z_i / \sum_i Z_i},$$

denominator $\hat{\pi}_c = 1 - \hat{\pi}_a - \hat{\pi}_n$. Standard errors by asymptotics or bootstrap. Without A1–A3 only nonparametric bounds remain (Chapter 5), often too wide.

7.2.3.7 The "fourth condition": homogeneity

Homogeneity makes the IV estimand the population ATE. Any one of these suffices:

(a) constant effect, $Y(1) - Y(0) = \beta_1$;

(b) equal average effects across Z, $\mathbb{E}[Y(1) - Y(0) \mid Z = 0] = \mathbb{E}[Y(1) - Y(0) \mid Z = 1]$;

(c) no effect modification by U, $\mathbb{E}[Y(1) - Y(0) \mid U]$ constant.

(a) implies (b) implies (c).

(a) is rarely plausible, and (b) and (c) are still strong; applied work mostly relies on monotonicity and the complier-only LATE (Hernán and Robins 2020, §§16.3–16.4).

7.2.3.8 Summary

- **Linear IV:** $\beta_1 = \mathrm{Cov}(Y, Z)/\mathrm{Cov}(X, Z)$; IV = ILS = 2SLS (just-identified); constant effect.
- **LATE:** under A1–A3, CACE = $\mathrm{ITT}_y/\mathrm{ITT}_x$; heterogeneity allowed, local estimand.
- Under constant effects, CACE = ATE = β_1.

7.2.4 Appendix: Candidate instruments in observational studies

Type	Example	Key concern
Genetic variant (Mendelian randomization)	ALDH2 polymorphism, alcohol metabolism	pleiotropy (direct effect on Y?)
Provider preference	physician's prescribing tendency	unmeasured patient factors drive referral
Geographic/temporal access	distance to facility, policy change date	correlated with SES, urban/rural

Only relevance is testable; exclusion and no confounding need subject-matter knowledge (*What If*, Fine Point 16.1).

7.2.5 Appendix: IV versus other causal methods

IV estimation	IPW / g-methods
Does not require conditional exchangeability $Y(x) \perp\!\!\!\perp X \mid W$; works with unmeasured confounding.	Requires conditional exchangeability (all confounders measured).
Sensitive to violations of exclusion/no confounding, with bias amplification.	Minor violations give minor bias (no amplification).
Identifies only a local effect (compliers) under monotonicity.	Identifies the population ATE; extends to time-varying treatments.

Prefer IV with likely unmeasured confounding, a strong instrument, dichotomous treatment, and plausible homogeneity or monotonicity (*What If*, §16.6).

7.2.6 Appendix: Practical issues with classic IV studies

Quarter of birth (Angrist and Krueger 1991): weak (QOB explains $< 0.1\%$ of schooling variation); many-instruments bias (180 instruments push 2SLS toward OLS; Bound et al. 1995 got similar results with random data); exclusion (season of birth may affect health, school readiness); exogeneity

(birth timing tracks family background; Buckles and Hungerman 2013). Still foundational for IV/LATE.

Proximity to college (Card 1995): exclusion (correlated with urban/rural status, SES, school quality); monotonicity (may not uniformly raise schooling); LATE applies to marginal, likely lower-income compliers, so LATE $\neq$ ATE (Carneiro et al. 2011).

7.2.7 Appendix: Weak instruments

Three problems. (1) Wide confidence intervals: weak Z–X association inflates $\text{Var}(\hat{\beta}_{1,\text{iv}})$.

(2) Bias amplification: in $\hat{\beta}_1 = \text{ITT}_y/\text{ITT}_x$ a small numerator bias (minor exclusion violation) is divided by a small denominator.

(3) Finite-sample bias toward OLS, even with a valid instrument. Rule of thumb: first-stage $F < 10$ (test of $H_0 : \alpha = 0$ in $X = \alpha_0 + \alpha Z + \epsilon$) signals weakness (Staiger and Stock 1997; *What If*, §16.5).

Example (*What If*, §16.2): smoking cessation X, weight gain Y, cigarette price in birth state Z. $P(X = 1 \mid Z = 1) - P(X = 1 \mid Z = 0) = 25.8\% - 19.5\% = 6.3\%$, $F = 0.8 \ll 10$; IV estimate 2.4 kg, 95% CI $[-36.5, 41.3]$: uninformative.

What to do. Report the first-stage F against the threshold for K instruments (Stock and Yogo 2005). Use weak-instrument-robust sets: the Anderson–Rubin set bypasses the first stage and is optimal with one instrument. With many instruments prefer LIML to 2SLS (less biased) (Anderson and Rubin 1949; Andrews, Stock and Sun 2019; Montiel Olea and Pflueger 2013).

What not to do. Screen on $F > 10$ and report only passing specifications (worsens size distortion); shop for the instrument set with highest F; bootstrap IV inference; trust $\hat{\beta} \pm 1.96\,\text{SE}$ when F is small. The real fix is a stronger instrument.

7.2.8 Appendix: Many instruments and robust IV

Each of K instruments gives $\hat{\beta}_k = \text{Cov}(Y, Z_k)/\,\text{Cov}(X, Z_k)$; valid ones agree, invalid ones (e.g. pleiotropy) diverge and contaminate 2SLS. Robust alternatives: weighted median (Bowden et al. 2016), consistent if over 50% are valid; mode-based (Hartwig et al. 2017), consistent if a plurality agree; MR-Egger (Bowden et al. 2015), an intercept absorbing directional pleiotropy. Standard in Mendelian randomization with hundreds of genetic variants.